

Digital Logic (DL)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 64

Question 1 of 64 | GATE 2025 Set 1 (CS1) | Q.17 (Computer Science & IT)

Q.17

$g(\cdot)$ is a function from A to B , $f(\cdot)$ is a function from B to C , and their composition defined as $f(g(\cdot))$ is a mapping from A to C .

If $f(\cdot)$ and $f(g(\cdot))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(\cdot)$?

- (A) $g(\cdot)$ must be an onto (surjective) function.
- (B) $g(\cdot)$ must be a one-to-one (injective) function.
- (C) $g(\cdot)$ must be a bijective function, that is, both one-to-one and onto.
- (D) $g(\cdot)$ is not required to be a one-to-one or onto function.

Question 2 of 64 | GATE 2025 Set 1 (CS1) | Q.18 (Computer Science & IT)

Q.18

Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G . For any two vertices, u and v , let $d_1(u, v)$ and $d_2(u, v)$ be the shortest distances between u and v in G and T , respectively. Which ONE of the options is CORRECT for all possible G , T , u and v ?

- (A) $d_1(u, v) = d_2(u, v)$
- (B) $d_1(u, v) \leq d_2(u, v)$
- (C) $d_1(u, v) \geq d_2(u, v)$
- (D) $d_1(u, v) \neq d_2(u, v)$

Question 3 of 64 | GATE 2025 Set 1 (CS1) | Q.24 (Computer Science & IT)

Q.24

Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?

- (A) $X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$
- (B) $X(a, b, X(a, b, c)) = X(a, b, c)$
- (C) $X(a, b, X(a, c, d)) = (X(a, b, a) \text{ AND } X(c, d, c))$
- (D) $X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

Question 4 of 64 | GATE 2025 Set 1 (CS1) | Q.35 (Computer Science & IT)

Q.35

The height of any rooted tree is defined as the maximum number of edges in the path from the root node to any leaf node.

Suppose a Min-Heap T stores 32 keys. The height of T is _____.

(Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 5 of 64 | GATE 2025 Set 1 (CS1) | Q.37 (Computer Science & IT)

Q.37

A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction?

ADD R1, #25 // $R1 = R1 + 25$

- (A) 16
- (B) 20
- (C) 22
- (D) 24

Question 6 of 64 | GATE 2025 Set 1 (CS1) | Q.47 (Computer Science & IT)

Q.47

Consider a relational schema $team(name, city, owner)$, with functional dependencies $\{name \rightarrow city, name \rightarrow owner\}$.

The relation $team$ is decomposed into two relations, $t1(name, city)$ and $t2(name, owner)$. Which of the following statement(s) is/are TRUE?

- (A) The relation $team$ is NOT in BCNF.
- (B) The relations $t1$ and $t2$ are in BCNF.
- (C) The decomposition constitutes a lossless join.
- (D) The relation $team$ is NOT in 3NF.

Question 7 of 64 | GATE 2025 Set 1 (CS1) | Q.51 (Computer Science & IT)

Q.51

A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, $1K = 2^{10}$ and $1M = 2^{20}$. The amount of space in bytes

that will be wasted due to internal fragmentation is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 8 of 64 | GATE 2025 Set 1 (CS1) | Q.57 (Computer Science & IT)

Q.57

Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header.

The number of fragments that will be delivered to the destination is _____ .

(Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 9 of 64 | GATE 2025 Set 1 (CS1) | Q.59 (Computer Science & IT)

Q.59

Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer)

Present state

Next state

Output f

X= 0 X= 1 X= 0 X= 1

A

F

B

0

0

B

D

C

0

0

C

F

E

0

0

D

G

A

1

0

E
D
C
0
0
F
F
B
1
1
G
G
H
0
1
H
G
A
1
0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 10 of 64 | GATE 2025 Set 1 (CS1) | Q.62 (Computer Science & IT)

Q.62 Let LIST be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
    int data;
    struct list *next;
} LIST;
```

Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes.

Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____ .

(Answer in integer)

```
int find (int query, LIST *list) {
    while (list != NULL){
        if(list->data == query) return 1;
        list = list->next;
    }
    return 0;
}

int main () {
```

```
... ..  
ptr1=L1; ptr2=L2;  
while (ptr1->next != NULL){  
    query = ptr1->next->data;  
    if (find (query, L2))  
        ptr1->next = ptr1->next->next;  
    else ptr1 = ptr1->next;  
}  
... ..  
return 0;  
}
```

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 11 of 64 | GATE 2025 Set 2 (CS2) | Q.18 (Computer Science & IT)

Q.18

A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X.

Which ONE of the following is NOT a possible candidate for X?

- (A) Internet Control Message Protocol (ICMP)
- (B) Internet Group Management Protocol (IGMP)
- (C) Open Shortest Path First (OSPF)
- (D) Routing Information Protocol (RIP)

Question 12 of 64 | GATE 2025 Set 2 (CS2) | Q.34 (Computer Science & IT)

Q.34

In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 13 of 64 | GATE 2025 Set 2 (CS2) | Q.36 (Computer Science & IT)

Q.36

Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?

- (A) 88.23
- (B) 93.75
- (C) 85.44
- (D) 66.67

Question 14 of 64 | GATE 2025 Set 2 (CS2) | Q.41 (Computer Science & IT)

Q.41

An array A of length n with distinct elements is said to be bitonic if there is an index $1 \leq i \leq n$ such that $A[1..i]$ is sorted in the non-decreasing order and $A[i+1..n]$ is sorted in the non-increasing order.

Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A ?

- (A) $\Theta(n)$
- (B) $\Theta(1)$
- (C) $\Theta(\log^2 n)$
- (D) $\Theta(\log n)$

Question 15 of 64 | GATE 2025 Set 2 (CS2) | Q.43 (Computer Science & IT)

Q.43

Given the following Karnaugh Map for a Boolean function $F(w, x, y, z)$:

Which one or more of the following Boolean expression(s) represent(s) F ?

- (A) $\overline{w}x\overline{y}z + w\overline{x}\overline{y}z + \overline{w}x\overline{y}\overline{z} + w\overline{x}\overline{y}\overline{z} + xz$
- (B) $\overline{w}x\overline{y}z + \overline{w}x\overline{y}\overline{z} + w\overline{x}\overline{y}z + xz$
- (C) $\overline{w}x\overline{y}z + w\overline{x}\overline{y}z + w\overline{x}\overline{y}\overline{z} + xz$
- (D) $\overline{x}z + xz$

Question 16 of 64 | GATE 2024 Set 1 (CS1 S5) | Q.25 (Computer Science & IT)

Q.25

Which of the following process state transitions is/are NOT possible?

- (A) Running to Ready
- (B) Waiting to Running
- (C) Ready to Waiting
- (D) Running to Terminated

Question 17 of 64 | GATE 2024 Set 1 (CS1 S5) | Q.64 (Computer Science & IT)

Q.64

Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2.

A, B, and C are select lines of M1, M2, and M3, respectively.

M1

M2

M3

A

B

C

1

0

1

0

1

0

X2

X1

X4

X3

Y

Q1

Q2

S1

S2

S3

For an instance of inputs $X1=1$, $X2=1$, $X3=0$, and $X4=0$, the number of combinations of A, B, C that give the output $Y=1$ is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 18 of 64 | GATE 2024 Set 2 (CS2 S6) | Q.51 (Computer Science & IT)

Q.51

Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has even weight. Which of the following statements is/are TRUE for every such graph G ?

- (A) All edges in G have even weight
- (B) All edges in G have even weight OR all edges in G have odd weight
- (C) In each cycle C in G, all edges in C have even weight

- (D) In each cycle C in G , either all edges in C have even weight OR all edges in C have odd weight

Question 19 of 64 | GATE 2024 Set 2 (CS2 S6) | Q.56 (Computer Science & IT)

Q.56

A functional dependency $F: X \rightarrow Y$ is termed as a useful functional dependency if and only if it satisfies all the following three conditions:

- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 64 | GATE 2023 Single Session | Q.16 (Computer Science & IT)

Q.16

Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

- (A) Longer link length and lower transmission rate
- (B) Longer link length and higher transmission rate
- (C) Shorter link length and lower transmission rate
- (D) Shorter link length and higher transmission rate CS

Question 21 of 64 | GATE 2023 Single Session | Q.19 (Computer Science & IT)

Q.19

CS

An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions.

Let k be the number of keys, m be the number of slots in the hash table, and $k > m$.

Which one of the following is the best hashing strategy to counteract the adversary?

- (A) Division method, i.e., use the hash function $h(k) = k \bmod m$.
- (B) Multiplication method, i.e., use the hash function $h(k) = \lfloor m(kA - \lfloor kA \rfloor) \rfloor$, where A is a carefully chosen constant.
- (C) Universal hashing method.
- (D) If k is a prime number, use Division method. Otherwise, use Multiplication method. CS

Question 22 of 64 | GATE 2023 Single Session | Q.20 (Computer Science & IT)

Q.20

The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.

0

1

S

Q

Multiplexer

Match the functional equivalence of this circuit to one of the following options.

- (A) D Flip-flop
- (B) D Latch
- (C) Half-adder
- (D) Demultiplexer CS

Question 23 of 64 | GATE 2023 Single Session | Q.40 (Computer Science & IT)

Q.40

Consider the given C-code and its corresponding assembly code, with a few operands U1–U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code.

The memory is byte-addressable.

//C-code

int a[10], b[10], i;

// int is 32-bit

for (i=0; i<10;i++)

a[i] = b[i] * 8;

;assembly-code (; indicates comments)

;r1-r5 are 32-bit integer registers

;initialize r1=0, r2=10

;initialize r3, r4 with base address of a, b

L01: jeq r1, r2, end

;if(r1==r2) goto end

L02: lw

r5, 0(r4)

;r5 <- Memory[r4+0]

L03: shl r5, r5, U1

;r5 <- r5 << U1

L04: sw

r5, 0(r3)

;Memory[r3+0] <- r5

L05: add r3, r3, U2

;r3 <- r3+U2

L06: add r4, r4, U3

L07: add r1, r1, 1

L08: jmp U4

;goto U4

L09: end

Which one of the following options is a CORRECT replacement for operands in the position (U1, U2, U3, U4) in the above assembly code?

- (A) (8, 4, 1, L02)
- (B) (3, 4, 4, L01)
- (C) (8, 1, 1, L02)
- (D) (3, 1, 1, L01) CS

Question 24 of 64 | GATE 2023 Single Session | Q.41 (Computer Science & IT)

Q.41

A 4 kilobyte (KB) byte-addressable memory is realized using four 1 KB memory blocks. Two input address lines (IA4 and IA3) are connected to the chip select (CS) port of these memory blocks through a decoder as shown in the figure. The remaining ten input address lines from IA11–IA0 are connected to the address port of these blocks. The chip select (CS) is active high.

X1

X2

X3

X4

Decoder

1KB memory

1KB memory

1KB memory

1KB memory

Q0

Q1

Q2

Q3

A1

A0

CS

CS

CS

CS

IA11

IA10

IA9

IA8
 IA7
 IA6
 IA5
 IA2
 IA1
 IA0
 IA4
 IA3
 Addr
 Addr
 Addr
 Addr
 10-bit
 MSB
 LSB

The input memory addresses (IA11–IA0), in decimal, for the starting locations (Addr=0) of each block (indicated as X1, X2, X3, X4 in the figure) are among the options given below. Which one of the following options is CORRECT?

- (A) (0, 1, 2, 3)
- (B) (0, 1024, 2048, 3072)
- (C) (0, 8, 16, 24)
- (D) (0, 0, 0, 0) CS

Question 25 of 64 | GATE 2023 Single Session | Q.43 (Computer Science & IT)

Q.43

A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0–X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.

X0
 X1
 X2
 X3
 A
 C
 X4
 X5
 X6
 X7
 A

C
 B
 0
 1
 2
 3
 0
 1
 2
 3
 S1 S0
 S1 S0
 S
 0
 1
 Multiplexer
 Multiplexer
 Multiplexer
 Q
 Q
 Q
 M1
 M2
 M3

Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function $A + A.C + A.B.C$?

- (A) (1, 1, 0, 0, 1, 1, 1, 0)
- (B) (1, 1, 0, 0, 1, 1, 0, 1)
- (C) (1, 1, 0, 1, 1, 1, 0, 0)
- (D) (0, 0, 1, 1, 0, 1, 1, 1) CS

Question 26 of 64 | GATE 2023 Single Session | Q.45 (Computer Science & IT)

Q.45

Let A be a priority queue for maintaining a set of elements.

Suppose A is

implemented using a max-heap data structure. The operation Extract-Max(A) extracts and deletes the maximum element from A. The operation Insert(A, key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations.

When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?

- (A) Both Extract-Max(A) and Insert(A,key) run in $O(1)$.
- (B) Both Extract-Max(A) and Insert(A,key) run in $O(\log(n))$.
- (C) Extract-Max(A) runs in $O(1)$ whereas Insert(A,key) runs in $O(n)$.
- (D) Extract-Max(A) runs in $O(1)$ whereas Insert(A,key) runs in $O(\log(n))$. CS

Question 27 of 64 | GATE 2023 Single Session | Q.47 (Computer Science & IT)

Q.47

Let $U = \{1, 2, \dots, n\}$, where n is a large positive integer greater than 1000. Let k be a positive integer less than n . Let A, B be subsets of U with $|A| = |B| = k$ and $A \cap B = \emptyset$. We say that a permutation of U separates A from B if one of the following is true.

- All members of A appear in the permutation before any of the members of B .
- All members of B appear in the permutation before any of the members of A .

How many permutations of U separate A from B ?

- (A) $n!$
- (B) $\frac{n!}{2^k} \cdot \frac{(n-2k)!}{(k!)^2}$
- (C) $\frac{n!}{2^k} \cdot \frac{(n-2k)!}{(k!)^2}$
- (D) $2 \cdot \frac{n!}{2^k} \cdot \frac{(n-2k)!}{(k!)^2}$ CS

Question 28 of 64 | GATE 2023 Single Session | Q.49 (Computer Science & IT)

Q.49

Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

- (A) 30 bits
- (B) 32 bits
- (C) 34 bits
- (D) 36 bits CS

Question 29 of 64 | GATE 2023 Single Session | Q.54 (Computer Science & IT)

Q.54

Let G be a simple, finite, undirected graph with vertex set $\{v_1, \dots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $N = \{1, 2, \dots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy:

for $i = 1, \dots, n$

color(v_i) $\leftarrow \min\{j \in N :$

no neighbour of v_i is colored $j\}$

Which of the following statements is/are TRUE?

- (A) This procedure results in a proper vertex coloring of G .
- (B) The number of colors used is at most $\Delta(G) + 1$.
- (C) The number of colors used is at most $\Delta(G)$.
- (D) The number of colors used is equal to the chromatic number of G . CS

Question 30 of 64 | GATE 2023 Single Session | Q.55 (Computer Science & IT)

Q.55

Let $U = \{1, 2, 3\}$. Let $2U$ denote the powerset of U . Consider an undirected graph

G whose vertex set is $2U$. For any $A, B \in 2U$, (A, B) is an edge in G if and only

if (i) $A \neq B$, and (ii) either $A \subseteq B$ or $B \subseteq A$. For any vertex A in G , the set of

all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by $B(A)$.

If $\emptyset$ denotes the empty set, then the cardinality of $B(\emptyset)$ is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 31 of 64 | GATE 2023 Single Session | Q.61 (Computer Science & IT)

Q.61

Consider a database of fixed-length records, stored as an ordered file. The database has 25,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.

(15 Bytes)

Primary Key

Other Fields

(85 Bytes)

Data File

Primary Key

Block Anchor

Pointer

Block

Index File

Suppose the block size of the file system is 1024 bytes, and a pointer to a block

occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.

Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 32 of 64 | GATE 2022 Single Session | Q.11 (Computer Science & IT)

Q.11

Which one of the following statements is TRUE for all positive functions

()

$f(n)$?

- (A) $2^{\lceil \log_2(n) \rceil} \leq f(n) \leq 2^{\lfloor \log_2(n) \rfloor}$ when $f(n)$ is a polynomial
- (B) $2^{\lceil \log_2(n) \rceil} \leq f(n) \leq 2^{\lfloor \log_2(n) \rfloor}$
- (C) $2^{\lceil \log_2(n) \rceil} \leq f(n) \leq 2^{\lfloor \log_2(n) \rfloor}$ when $f(n)$ is an exponential function
- (D) $2^{\lceil \log_2(n) \rceil} \leq f(n) \leq 2^{\lfloor \log_2(n) \rfloor}$

Question 33 of 64 | GATE 2022 Single Session | Q.14 (Computer Science & IT)

Q.14

In a relational data model, which one of the following statements is TRUE?

- (A) A relation with only two attributes is always in BCNF.
- (B) If all attributes of a relation are prime attributes, then the relation is in BCNF.
- (C) Every relation has at least one non-prime attribute.
- (D) BCNF decompositions preserve functional dependencies.

Question 34 of 64 | GATE 2022 Single Session | Q.16 (Computer Science & IT)

Q.16

Suppose we are given n keys, m hash table slots, and two simple uniform hash functions

h_1 and

h_2 . Further suppose our hashing scheme uses

h_1 for the odd keys

and

h_2 for the even keys. What is the expected number of keys in a slot?

- (A) m/n

- (B) $n m$
- (C) $2n m$
- (D) $2 n m$

Question 35 of 64 | GATE 2022 Single Session | Q.40 (Computer Science & IT)

Q.40

Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T?

- (A) P is 10:1 multiplexer; Q is 5:1 multiplexer; T is 2:1 multiplexer
- (B) P is 10:210 decoder; Q is 5:25 decoder; T is 2:1 encoder
- (C) P is 10:210 decoder; Q is 5:25 decoder; T is 2:1 multiplexer
- (D) P is 1:10 de-multiplexer; Q is 1:5 de-multiplexer; T is 2:1 multiplexer
X P Q T S R DDS M S-address R-address
Code Word

Question 36 of 64 | GATE 2022 Single Session | Q.49 (Computer Science & IT)

Q.49

Consider a simple undirected weighted graph

G all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?

- (A) The edge with the second smallest weight is always part of any minimum spanning tree of G .
- (B) One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G .
- (C) Suppose $S \subseteq V$ be such that $S \neq \emptyset$ and $S \neq V$. Consider the edge with the minimum weight such that one of its vertices is in S and the other in $V \setminus S$. Such an edge will always be part of any minimum spanning tree of G .
- (D) G can have multiple minimum spanning trees.

Question 37 of 64 | GATE 2021 Set 1 (CS1) | Q.6 (Computer Science & IT)

Q.6

We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm x 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the

largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____

- (A) π^2
- (B) 3π
- (C) 9π
- (D) 3π

Question 38 of 64 | GATE 2021 Set 1 (CS1) | Q.10 (Computer Science & IT)

Q.10

Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality.

Which one of the following statements summarizes the passage?

- (A) The proposed AOM addresses the core problems that cause obesity.
- (B) If obesity reduces, poverty will naturally reduce, since obesity causes poverty.
- (C) AOM are addressing the core problems and are likely to succeed.
- (D) AOM are addressing the problem superficially. Q.1 – Q.10 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Question 39 of 64 | GATE 2021 Set 2 (CS2) | Q.6 (Computer Science & IT)

Q.6

Listening to music during exercise improves exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music.

Which one of the following statements is the CORRECT inference of the above passage?

- (A) Listening to music has no effect on learning and a positive effect on physical exercise.
- (B) Listening to music has a clear positive effect both on physical exercise and on learning.
- (C) Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students.
- (D) Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise.

Question 40 of 64 | GATE 2019 Single Session | Q.10 (Computer Science & IT)

Q.10 Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.

Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.

Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell.

Request by Z: I believe in Vaastu and so want to stay in the South-west wing.

The shaded rooms are already occupied. WR is washroom.

- (A) WR Garden N E Z Entrance W S Garden WR X Y
 - (B) WR X Garden N E Z Entrance W S Y Garden WR
 - (C) WR Z Garden N E Entrance W S Y Garden WR X
 - (D) WR Garden N E Entrance W S Y Garden WR X Z
- END OF THE QUESTION PAPER CS 1/15

Question 41 of 64 | GATE 2019 Single Session | Q.26 (Computer Science & IT)

Q.26 Consider the following C function.

```
void convert(int n){
    if(n<0)
        printf("%d",n);
    else {
        convert(n/2);
        printf("%d",n%2);
    }
}
```

Which one of the following will happen when the function convert is called with any positive integer n as argument?

- (A) It will print the binary representation of n and terminate
- (B) It will print the binary representation of n in the reverse order and terminate
- (C) It will print the binary representation of n but will not terminate
- (D) It will not print anything and will not terminate

Question 42 of 64 | GATE 2019 Single Session | Q.29 (Computer Science & IT)

Q.29 Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

- (A) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X
- (B) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
- (C) X sends an ARP request packet with broadcast MAC address in its local subnet

- (D) X sends an ARP request packet with broadcast IP address in its local subnet

Question 43 of 64 | GATE 2019 Single Session | Q.30 (Computer Science & IT)

Q.30 Consider three 4-variable functions f_1 , f_2 , and f_3 , which are expressed in sum-of-minterms as

$$f_1 = \sum (0, 2, 5, 8, 14),$$

$$f_2 = \sum (2, 3, 6, 8, 14, 15),$$

$$f_3 = \sum (2, 7, 11, 14)$$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:

- (A) $\sum (7, 8, 11)$
 (B) $\sum (2, 7, 8, 11, 14)$
 (C) $\sum (2, 14)$
 (D) $\sum (0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

Question 44 of 64 | GATE 2019 Single Session | Q.32 (Computer Science & IT)

Q.32 Let the set of functional dependencies $F = \{QR \rightarrow S, R \rightarrow P, S \rightarrow Q\}$ hold on a relation schema $X = (PQRS)$. X is not in BCNF. Suppose X is decomposed into two schemas Y and Z , where $Y = (PR)$ and $Z = (QRS)$.

Consider the two statements given below.

I.

Both Y and Z are in BCNF

II.

Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

- (A) Both I and II
 (B) I only
 (C) II only
 (D) Neither I nor II

Question 45 of 64 | GATE 2019 Single Session | Q.35 (Computer Science & IT)

Q.35 Consider the first order predicate formula ϕ :

$$\exists x [(\exists z x|z \wedge ((z = x) \vee (z = 1))) \wedge \exists w (w > x) \wedge (\exists z z|w \wedge ((w = z) \vee (z = 1)))]$$

Here ' $a|b$ ' denotes that ' a divides b ', where a and b are integers. Consider the following sets:

S1.

$$\{1, 2, 3, \dots, 100\}$$

S2.

Set of all positive integers

S3.

Set of all integers

Which of the above sets satisfy φ ?

- (A) S1 and S2
- (B) S1 and S3
- (C) S2 and S3
- (D) S1, S2 and S3 CS 9/15

Question 46 of 64 | GATE 2019 Single Session | Q.50 (Computer Science & IT)

Q.50 What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as $f = \Sigma (0, 2, 5, 7, 8, 10, 13, 15)$? Assume that all the inputs and their complements are available. Answer: _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 47 of 64 | GATE 2018 Single Session | Q.11 (Computer Science & IT)

Q.11 In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in R and that the cardinality of E1 is greater than the cardinality of E2.

Which one of the following is true about R?

- (A) Every entity in E1 is associated with exactly one entity in E2.
- (B) Some entity in E1 is associated with more than one entity in E2.
- (C) Every entity in E2 is associated with exactly one entity in E1.
- (D) Every entity in E2 is associated with at most one entity in E1. CS 4/20

Question 48 of 64 | GATE 2018 Single Session | Q.19 (Computer Science & IT)

Q.19 Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 49 of 64 | GATE 2018 Single Session | Q.20 (Computer Science & IT)

Q.20 The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 50 of 64 | GATE 2018 Single Session | Q.21 (Computer Science & IT)

Q.21 Consider the following C program:

```
#include <stdio.h>
int counter = 0;
int calc (int a, int b) {
    int c;
    counter++;
    if (b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}
int main (){
    calc(4, 81);
    printf ("%d", counter);
}
```

The output of this program is ____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 51 of 64 | GATE 2018 Single Session | Q.27 (Computer Science & IT)

Q.27 Let N be the set of natural numbers. Consider the following sets.

P: Set of Rational numbers (positive and negative)

Q: Set of functions from $\{0, 1\}$ to N

R: Set of functions from N to $\{0, 1\}$

S: Set of finite subsets of N .

Which of the sets above are countable?

- (A) Q and S only
- (B) P and S only
- (C) P and R only
- (D) P, Q and S only

Question 52 of 64 | GATE 2018 Single Session | Q.28 (Computer Science & IT)

Q.28 Consider the first-order logic sentence

$\varphi \equiv \exists s \exists t \exists u \exists v \exists w \exists x \exists y \psi(s, t, u, v, w, x, y)$

where $\psi(s, t, u, v, w, x, y)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

- (A) There exists at least one model of φ with universe of size less than or equal to 3.
- (B) There exists no model of φ with universe of size less than or equal to 3.
- (C) There exists no model of φ with universe of size greater than 7.
- (D) Every model of φ has a universe of size equal to 7. CS 9/20

Question 53 of 64 | GATE 2018 Single Session | Q.33 (Computer Science & IT)

Q.33 Consider the unsigned 8-bit fixed point binary number representation below,
b7 b6 b5 b4 b3 . b2 b1 b0

where the position of the binary point is between b3 and b2. Assume b7 is the most significant bit. Some of the decimal numbers listed below cannot be represented exactly in the above representation:

- (i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true?

- (A) None of (i), (ii), (iii), (iv) can be exactly represented
- (B) Only (ii) cannot be exactly represented
- (C) Only (iii) and (iv) cannot be exactly represented
- (D) Only (i) and (ii) cannot be exactly represented CS 11/20

Question 54 of 64 | GATE 2018 Single Session | Q.49 (Computer Science & IT)

Q.49 Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is ____.

x

5

3

4

1

4

4

CS

18/20

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 55 of 64 | GATE 2018 Single Session | Q.55 (Computer Science & IT)

Q.55 Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following

carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance d meters from each other. P starts transmitting a packet at time $t=0$ after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time $t=0$ and begins to carrier-sense the medium. The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is ____.

END OF THE QUESTION PAPER

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 56 of 64 | GATE 2016 Set 1 (CS1) | Q.1 (Computer Science & IT)

Q.1

Let p, q, r, s represent the following propositions.

$p: x \in \{8, 9, 10, 11, 12\}$

$q: x$ is a composite number

$r: x$ is a perfect square

$s: x$ is a prime number

The integer $x \geq 2$ which satisfies $\neg((p \wedge q) \wedge (\neg r \wedge \neg s))$ is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 57 of 64 | GATE 2016 Set 1 (CS1) | Q.8 (Computer Science & IT)

Q.8

We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats.

The minimum number of J-K flip-flops required to implement this counter is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 58 of 64 | GATE 2016 Set 1 (CS1) | Q.30 (Computer Science & IT)

Q.30

Consider the two cascaded 2-to-1 multiplexers as shown in the figure.

MUX

2-to-1

0

1

0

1

0

MUX

2-to-1

X

R

R

P

Q

s

s

The minimal sum of products form of the output X is

(A) $P'Q + PQR$

(B) $PQ + QR$

(C) $PQ + P'QR$

(D) $Q'R + PQR$

Question 59 of 64 | GATE 2016 Set 1 (CS1) | Q.39 (Computer Science & IT)

Q.39

Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of G can have is

.

CS(Set A)

12/17

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 60 of 64 | GATE 2016 Set 1 (CS1) | Q.41 (Computer Science & IT)

Q.41

Let Q denote a queue containing sixteen numbers and S be an empty stack. Head(Q) returns the element at the head of the queue Q without removing it from Q. Similarly Top(S) returns the element at the top of S without removing it from S. Consider the algorithm given below.

while Q is not Empty do

if S is Empty OR $\text{Top}(S) \leq \text{Head}(Q)$ then

x := Dequeue(Q);

```
Push(S,x);  
else  
x := Pop(S);  
Enqueue(Q,x);  
end  
end
```

The maximum possible number of iterations of the while loop in the algorithm is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 61 of 64 | GATE 2016 Set 2 (CS2) | Q.10 (Computer Science & IT)

Q.10

A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is

.

CS(Set B)

2/18

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 62 of 64 | GATE 2016 Set 2 (CS2) | Q.26 (Computer Science & IT)

Q.26

A binary relation R on $N \times N$ is defined as follows: $(a,b)R(c,d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

P: R is reflexive

Q: R is transitive

Which one of the following statements is TRUE?

- (A) Both P and Q are true.
- (B) P is true and Q is false.
- (C) P is false and Q is true.
- (D) Both P and Q are false.

Question 63 of 64 | GATE 2016 Set 2 (CS2) | Q.30 (Computer Science & IT)

Q.30

Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units UF and UG , respectively. Given two instances of UF and two instances of UG , it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is

nanoseconds.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 64 of 64 | GATE 2016 Set 2 (CS2) | Q.35 (Computer Science & IT)

Q.35

The following function computes XY for positive integers X and Y .

```
int exp(int X, int Y) {  
    int res = 1, a = X, b = Y;  
    while ( b != 0 ){  
        if ( b%2 == 0 ) { a = a*a; b = b/2; }  
        else  
            { res = res*a; b = b-1; }  
    }  
    return res;  
}
```

Which one of the following conditions is TRUE before every iteration of the loop?

- (A) $XY = ab$
- (B) $(res \square a)Y = (res \square X)b$
- (C) $XY = res \square ab$
- (D) $XY = (res \square a)b$ CS(Set B) 9/18 GATE 2016